



## Technical Data Sheet

Previous Name: Shell Morlina Oils

# Shell Morlina S2 B 220

- *Reliable Protection*
- *Industrial Application*
- *Water Shedding*

## Industrial Bearing & Circulating Oils

Shell Morlina S2 B oils are high performance oils designed to provide outstanding oxidation and water separation protection for most general industrial bearing and circulating oil system applications and certain other industrial applications which do not require oils with extreme pressure (EP) properties. These oils meet the requirements of the Morgan Construction Company and Danieli for common bearing oils.

## DESIGNED TO MEET CHALLENGES

### Performance, Features & Benefits

#### ▪ Long oil life – Maintenance saving

Shell Morlina S2 B oils are formulated with a well proven rust and oxidation inhibitor additive package that helps provide consistent performance and protection throughout the maintenance interval.

#### ▪ Reliable wear & corrosion protection

Shell Morlina S2 B oils help prolong the life of bearings and circulating systems through:

- Excellent water separation characteristics that helps ensure that critical oil films are retained between highly loaded parts.
- Good air release characteristics to minimize cavitation and associated damage to circulating pumps.
- Helps protect against corrosion, oxidation, and emulsion formation, even in the presence of water.

#### ▪ Maintaining system efficiency

Shell Morlina S2 B oils are blended with high quality, solvent refined base oils that promote good water separation and air release to ensure the efficient lubrication of the machines and systems.

#### ▪ Roll-neck bearings

#### ▪ Enclosed industrial gear systems

Low or moderately loaded enclosed gears where EP performance is not required.

### Specifications, Approvals & Recommendations

- Morgan MORGOIL® Lubricant Specification New Oil (Rev. 1.1) (MORGOIL is a registered trademark of the Morgan Construction Company)
- Danieli Standard Oil 6.124249.F
- DIN 51517-1 – type C
- DIN 51517-2 - type CL

For a full listing of equipment approvals and recommendations, please consult your local Shell Technical Helpdesk, or the OEM Approvals website.

### Compatibility & Miscibility

#### ▪ Paint Compatibility

Shell Morlina S2 B oils are compatible with seal materials and paints normally specified for use with mineral oils.

### Main Applications



- Machine circulation systems
- Oil lubricated bearings

Suitable for most plain and rolling element bearings and general industrial applications.

## Typical Physical Characteristics

| Properties                                    |        |                      | Method     | Shell Morlina S2 B |
|-----------------------------------------------|--------|----------------------|------------|--------------------|
| ISO Viscosity Grade                           |        |                      | ISO 3448   | 220                |
| Kinematic Viscosity                           | @40°C  | mm <sup>2</sup> /s   | ASTM D445  | 220                |
| Kinematic Viscosity                           | @100°C | mm <sup>2</sup> /s   | ASTM D445  | 18.3               |
| Density                                       | @15°C  | kg/m <sup>3</sup>    | ISO 12185  | 891                |
| Viscosity Index                               |        |                      | ISO 2909   | 92                 |
| Flash Point (COC)                             |        | °C                   | ISO 2592   | 280                |
| Pour Point                                    |        | °C                   | ISO 3016   | -15                |
| Rust, Distilled Water                         |        |                      | ASTM D665A | Pass               |
| Emulsion Test - @82°C (Unless specified by *) |        | Mins                 | ASTM D1401 | 20                 |
| Oxidation Control Test : TOST                 |        | Hrs                  | ASTM D943  | 1300+              |
| Oxidation Control Test : RBOT                 |        | Mins                 | ASTM D2272 | 400+               |
| Foam Test, Seq II                             |        | ml foam at 0/10 mins | ASTM D892  | 10/0               |

These characteristics are typical of current production. Whilst future production will conform to Shell's specification, variations in these characteristics may occur. \*@54°C

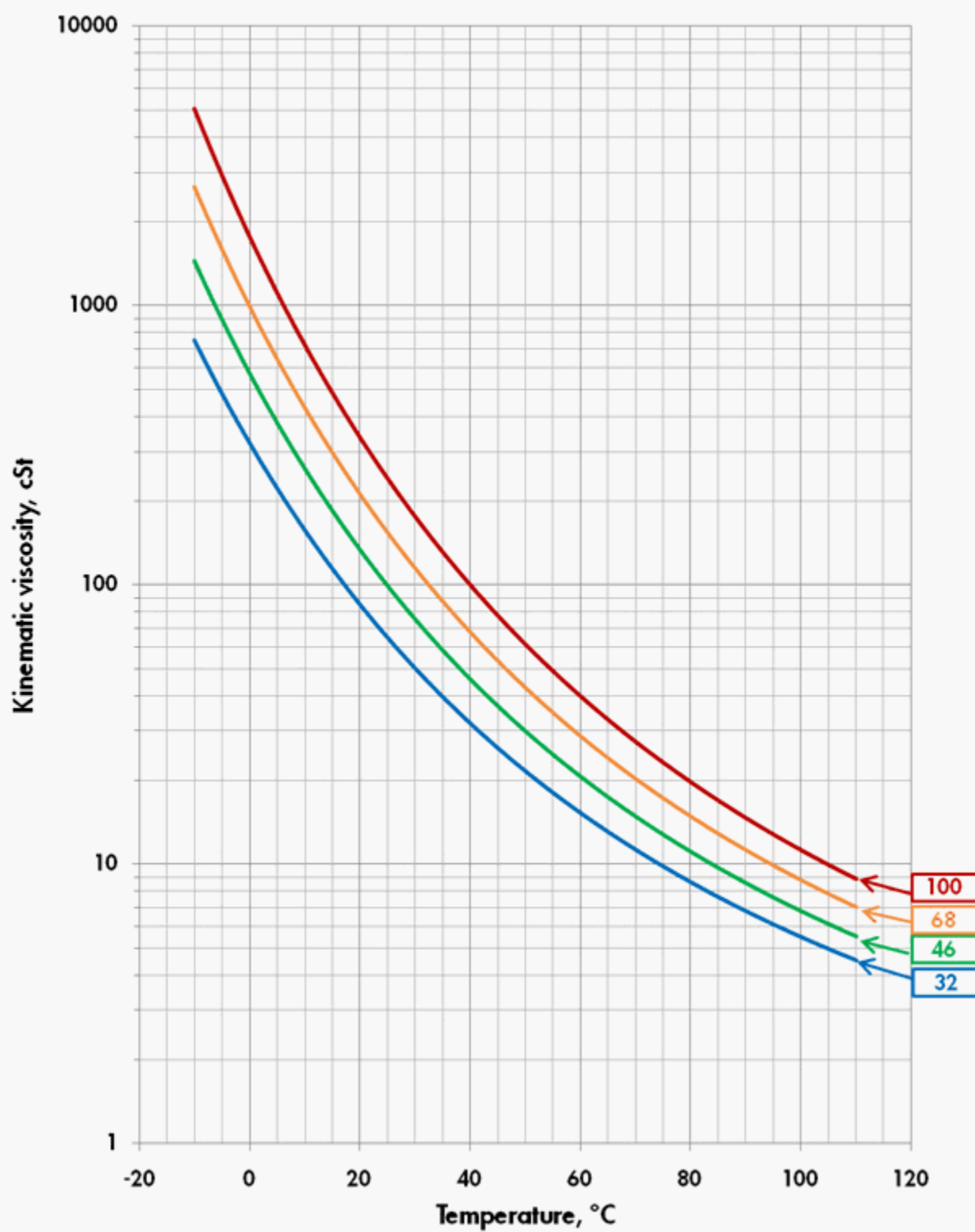
## Health, Safety & Environment

- Guidance on Health and Safety is available on the appropriate Material Safety Data Sheet, which can be obtained from <http://www.epc.shell.com/>
- **Protect the Environment**  
Take used oil to an authorised collection point. Do not discharge into drains, soil or water.

## Additional Information

- **Advice**  
Advice on applications not covered here may be obtained from your Shell representative.

Viscosity - Temperature Diagram for Shell Morlina S2 B



Viscosity - Temperature Diagram for Shell Morlina S2 B

